



Advancing Circular Economy of Bio-Based Building Materials: Critical Gaps and Future Prospects

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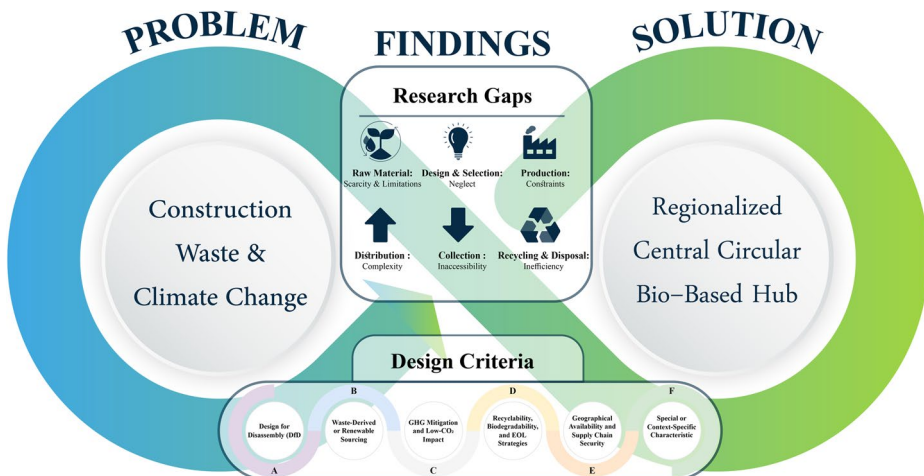
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Abstract

The construction industry is a significant contributor to climate change, exacerbating emissions and landfill waste in a world that has already exceeded the 1.5 °C threshold. In response, circular economy models and regenerative bio-based building materials have emerged as promising solutions. While bio-based materials align with the principles of the Bioeconomy, their integration into circular economy systems remains limited and underdeveloped, with recent studies failing to adequately identify and address key gaps. This paper employs a semi-systematic review methodology to critically examine all stages of the circular economy model for bio-based materials, categorized into early stages (sourcing, material design, and production) and end-of-life stages (collection and recycling). The review identifies 22 barriers and research gaps, which significantly affect end-of-life planning, collection infrastructure, and binder compatibility for recycling. A six-criteria framework for circular material design and selection is proposed. Non-wood materials such as bamboo, mycelium, and fiber-reinforced earth composites are identified as particularly promising, and the combined use of wood and non-wood bio-based materials is recommended to mitigate biodiversity loss and enhance resilience. Additionally, the study introduces a for-profit centralized intermediary hub model to coordinate the currently fragmented bio-based material markets by facilitating design decisions, supply chain integration, amid policy implementation. This review is the first to present a comprehensive circularity gap analysis, a circularity-focused material design framework, and a strategic intermediary model for bio-based construction materials, offering critical insights and actionable pathways for policymakers and industry stakeholders to support their mainstream adoption.

Extended author information available on the last page of the article

Graphical Abstract



Keywords Circular economy · Bio-Based materials · Construction · Building materials · EOL · Recycling

Abbreviations

BBMs	Bio-Based Building Materials
GHG	Greenhouse Gas
CDW	Construction and Demolition Waste
LCA	Life Cycle Assessment
GLT	Glued Laminated Timber
CBE	Circular Bioeconomy
CLT	Cross-Laminated Timber
DFD	Design for Disassembly
EOL	End of Life
CE	Circular Economy
LVL	Laminated Veneer Lumber

Introduction

In 2024, Earth exceeded the critical 1.5 °C warming threshold for the first time [1]. Accelerated by population growth, rising greenhouse gas (GHG) emissions, resource depletion, and fossil fuel use [2]. The intensification of climate change is disrupting weather systems and triggering sea-level rise, droughts, floods, dust storms, wildfires, and both ecological and human distress [3, 4]. With this threshold now breached, urgent action is needed to mitigate the escalating crisis. Unfortunately, the built environment is one of the primary sources of environmental strain. It exerts the most significant impact on the natural environment compared to all other industrial sectors [5], by contributing approximately 39% of global CO₂ emissions annually [6], consuming nearly 40% of raw materials [7, 8], and generat-

ing at least 30% of solid waste through construction and demolition waste (CDW) [9], with volumes expected to grow [10]. These impacts span the product life cycle: extraction, transport, production, use, recycling, and disposal [11]. Among these, the use phase, due to its operational energy and GHG emissions, has drawn the most research attention [12, 13], often overshadowing embodied impacts.

In a broader spectrum of the built environment, it has been reckoned that operational carbon continues to decline with technological advancements, embodied carbon remains constant, and it will surpass operational impact [14, 15]. This shift has led to policy responses, such as the low-embodied carbon program in the U.S. Inflation Reduction Act of 2022 [16], and UK regulations targeting embodied carbon [17], emphasizing the need for low-carbon construction solutions nationwide. Yet many studies rely on partial scopes, like “Cradle-to-Gate” or “Cradle-to-Grave”, often excluding the end-of-life (EOL) phase [18]. This oversight highlights the need to comprehensively address carbon emissions, encompassing recycling, reuse, and waste disposal.

In parallel, circular strategies for conventional construction materials, such as recycled concrete aggregates, reclaimed steel, and aluminum, have gained traction [19]. However, these processes often face limitations such as energy-intensive reprocessing, quality degradation through downcycling, and challenges of scrap quality [19, 20]. As a result, the circular potential of mineral and metal-based materials remains constrained and complex, reinforcing the need to explore materials with inherently renewable, low-carbon, and regenerative characteristics.

One multifaceted approach to mitigating GHG emissions throughout the life cycle, including EOL, is using bio-based building materials (BBMs) [21], which have been demonstrated to reduce emissions by an average of 45% across all material types [22]. In construction, BBMs refer to materials derived from plant waste, plant products, or animal biomass, used in both structural and decorative applications [21]. Examples include structural timber, insulation, paints, building blocks, and finishing elements [21, 23], sourced mainly from natural fibers like agricultural residue, timber, wood particles [21], and wool [21]. While BBMs have been used for millennia [24, 25], their environmental benefits, renewability, recyclability, porosity, compostability, and biodegradability have gained renewed interest amid climate concerns [26]. Lastly, unlike conventional materials such as concrete, which offer limited post-consumer options, BBMs support recycling, energy recovery, and reuse [27], positioning them as a promising environmental solution for the built environment.

From a framework perspective, the circular economy (CE) model provides a valuable approach to managing GHG emissions in BBMs and optimizing EOL phases by slowing, closing, and narrowing material loops [28]. By the principal definition, this system is designed to reduce waste and disposal by continuously reusing resources and materials, prioritizing sustainability, minimizing environmental impact, and extending the life cycle of byproducts [29, 30]. Notably, it has been stated that BBMs exhibit circular characteristics by repurposing waste byproducts and regenerative plant-based materials, contributing to the Circular Bioeconomy (CBE) [31], with both often regarded as complementary. Their biodegradability and recyclability make BBMs effective in reducing landfill waste and supporting a less resource-intensive [32].

Research gaps of this research stem from the frequent neglect and exclusion of EOL stages in life cycle assessment (LCA) [18, 33], often dismissed as negligible despite its waste management potential [34]. Similarly, studies on BBMs tend to focus on performance

testing while overlooking EOL considerations [21]. Furthermore, much of the current literature on BBMs' circularity, such as [33, 35, 36], recycles similar findings and focuses predominantly on production and distribution aspects of the CE, while neglecting core CE principles like raw material sourcing, collection, and especially EOL processes, including recycling, reuse, recovery, and disposal [37, 38]. These studies also lacked a structured CE framework capable of comprehensively analyzing all stages of the material life cycle. Although some findings provided valuable insights, they were limited in scope and insufficient to identify and address the fundamental gaps in the field.

Additionally, a lack of a clear decision-making framework for material design was identified in a previous study [39], and no comprehensive framework or set of material design and selection criteria was found in the broader literature. These deficiencies, compounded by the previously mentioned gaps, served as a key motivation for initiating this study. This research also raises a critical question: despite the long-standing availability of BBMs and the extensive body of research highlighting their potential to advance sustainable construction and circular economy principles, why have they not achieved comparable success to conventional building materials?

This study aims to bridge that gap by providing a comprehensive review and proposing a whole-life CE framework for bio-based materials. It addresses the lack of decision-making tools, the exclusion of EOL stages, and other overlooked phases of the CE stages, gaps that risk misleading sustainability claims and policy inefficiencies. By aligning the findings with global sustainability targets such as the Sustainable Development Goals and the EU Green Deal, this research offers a timely and policy-relevant contribution to the field. The primary aims of this study are as follows:

- To identify gaps and limitations in the BBMs' circularity across all CE stages.
- To highlight future opportunities and propose strategic pathways for advancing the circularity, production, and global adoption of BBMs.

The paper is structured as follows: (2) Methodology, explains the research approach; (3) Early-Stage CE Analysis, examines sustainable design, production, and application of BBMs; (4) EOL Analysis, explores collection, reuse, recycling, and disposal processes; (5) Discussion, identifies existing gaps, limitations, and offers recommendations; and (6) Conclusion, summarizes the key findings and contributions.

Methodology

A fully systematic review was not suitable for this study because the literature on BBMs and circularity is highly fragmented, multidisciplinary, and unevenly distributed across academic and grey sources. In such emerging fields, a semi-systematic (hybrid) review approach is recommended to enable both structured evidence retrieval and broader thematic integration. This method allows the review to capture a wide range of sources, such as conceptual, policy-based, and technical insights that would be excluded by strict systematic criteria [40].

Semi-systematic review has been considered more effective, as it enables broader thematic coverage and accommodates a wider range of topics than a traditional systematic

review [41]. Additionally, it provides a flexible conceptual framework for identifying knowledge gaps, emerging trends, and future research directions that traditional methods may overlook [41]. Given the fragmented and evolving nature of the field, snowballing and grey literature searches were used to identify additional relevant sources and improve comprehensiveness. Prior research has demonstrated that this approach is more effective in capturing relevant primary studies and preventing overstatement and recycling the outcomes due to emerging topics and a lack of knowledge in the field [42, 43].

This study follows a semi-systematic design that combines (a) a structured screening of peer-reviewed literature and (b) targeted narrative synthesis for contextualisation. The structured component was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist and flows [44]. As an evidence-based protocol, it offers methodological transparency and structured recommendations, distinguishing it from traditional systematic analyses [45]. The narrative component was necessary to interpret lifecycle gaps, design rationales, policy implications, and industry practices that are not consistently reported in database-indexed publications. This hybrid method follows established semi-systematic guidance for emerging research domains.

For data collection, Scopus, ScienceDirect, and Google Scholar databases were employed. We selected Scopus for its robustness and reliability as a citation database [46], and better content verification and quality than WOS [47]. Additionally, using Google Scholar, which offers unique citations and includes those found in WOS/Scopus, albeit with generally lower scientific impact [48], accompanied by ScienceDirect to access the full text. This study employed a manual content analysis technique, a scrutinizing and interpretive method for data examination [49–51]. The aim was to structure and interpret the collected data to derive meaningful, practical conclusions [52].

The keywords were chosen to reflect the research focus on early and EOL stages and to align closely with the study objectives. For exploratory keyword-based scoping, keywords were utilized in Google Scholar and ScienceDirect, including “construction,” “building,” “bio-based,” “timber,” “circular,” “circular economy,” “material,” “recycle,” “reuse,” “EOL,” “waste,” “upscaling,” “policy,” “challenges,” “barriers,” and “*variable building material*”, etc. The following query was used to search in the Scopus database for the systematic review: TITLE-ABS-KEY (bio-based) AND (construction OR building\$) AND material\$ AND (circular OR recycle OR “end-of-life” OR EOL). The search operation was conducted by focusing on the titles, abstracts, and keywords of articles, with no restrictions on the year of publication, thereby including all relevant literature available up to the search date.

A total of 232 ($n=232$) articles were initially retrieved. A five-year publication window was then applied to ensure that the review reflects the most recent developments in BBMs, as earlier studies generally lack contemporary circularity frameworks, EOL considerations, and updated sustainability assessment methods, reducing the pool to 213 studies. After applying standard screening filters (subject relevance, language, and quality), 72 articles remained for full-text assessment. Following full-text evaluation, 32 studies, supplied with additional articles aligned with the research objectives, were included in the final synthesis. Studies were excluded if they focused on non-construction applications of bio-based materials (e.g., medical, chemical, agricultural, or biofuel domains), as these fall outside the scope of BBMs in the built environment. Figure 1 provides a visual representation of the article selection and finalization process.

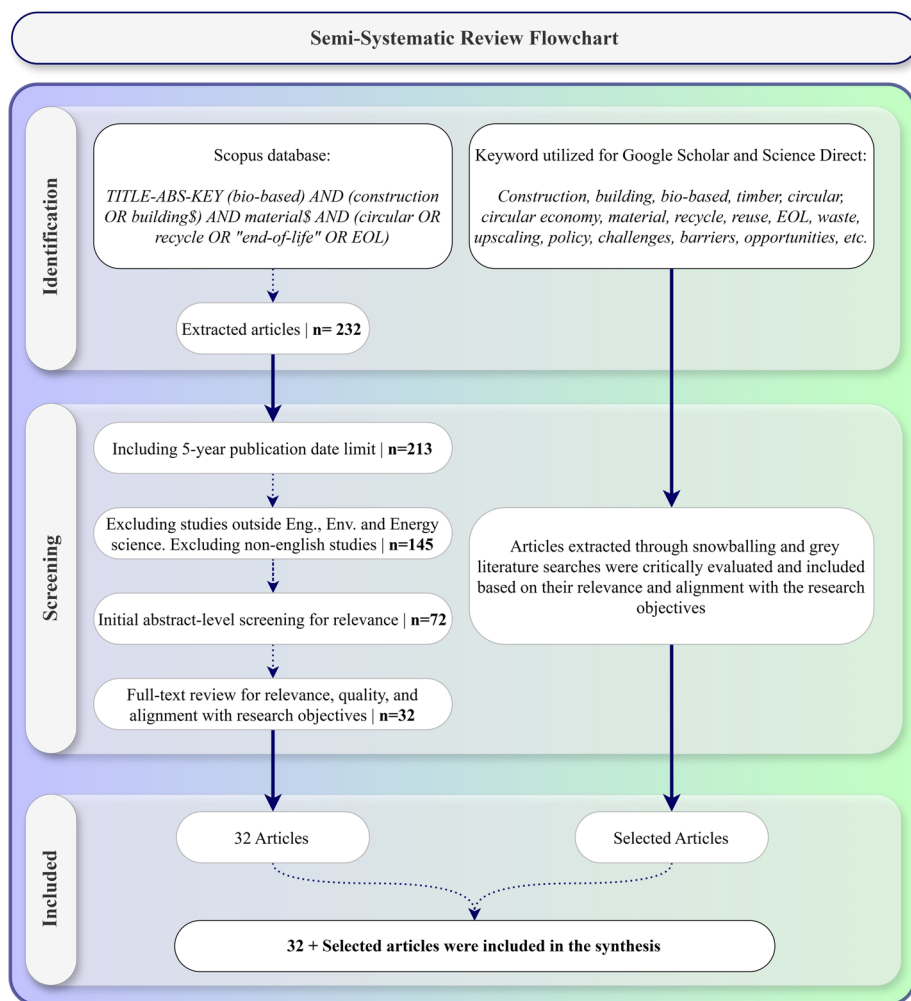


Fig. 1 Flowchart of literature collection and data cleansing process conducted through a semi-systematic review, following the PRISMA guidelines

Each Results subsection explicitly states whether it draws from:

- the 20 studies analysed for material design/selection criteria,
- the 32 studies included through structured screening, and,
- the additional literature incorporated through narrative synthesis.

The STEEP framework was initially considered as a thematic coding structure; however, the heterogeneity of the evidence produced overlapping and non-exclusive categories, reducing analytical clarity. For this reason, STEEP was not applied, and emphasis was placed instead on lifecycle-based classification aligned with CE stages.

This study adopts the comprehensive CE model proposed by [53], which is structured around seven stages: (1) raw materials, (2) sustainable design, (3) production, (4) distribution, (5) consumption, reuse, and repair, (6) collection, and (7) residual waste and waste management. For this analysis, the consumption, maintenance, repair, and use phases were excluded due to the high variability in material applications and the diverse hygrothermal and physical properties of BBMs, which render the analysis of their repair and maintenance highly complex. Figure 2 illustrates the scope of the study, which is categorized into two clustered stages as follows:

- 1 Early stage: This stage encompasses raw material selection, sustainable design, production, and distribution, phases that collectively determine over 80% of a product's total environmental impact [30]. Despite their pivotal role in shifting away from the traditional linear model, defined by the “take, make, and dispose” approach [29, 54], early-stage processes are often critically overlooked. Recognizing the early stage as the foundation for achieving circularity is essential, as it facilitates a comprehensive lifecycle assessment and proactive design interventions. This study examines how BBMs are selected, designed, and prepared for market integration, with a particular focus on renewability, availability, sustainability, and coherence with CE principles. Special emphasis is placed on design rationale and its influence on downstream circular outcomes, particularly regarding recyclability and potential for reuse.
- 2 EOL stage: The EOL stage includes reuse, repair, collection, residual waste, and waste management. This section examines how BBMs are disassembled from building components during demolition and evaluates the ease and efficiency of material collection. It further explores recycling and biodegradability potentials, focusing on recent advancements aimed at reducing environmental impacts during this phase. Additionally, disposal practices are reviewed to assess how effectively BBMs are diverted from landfills, repurposed, or reintegrated into material cycles. These processes are evaluated for their alignment with CE principles. By comparing the circularity of different BBMs,

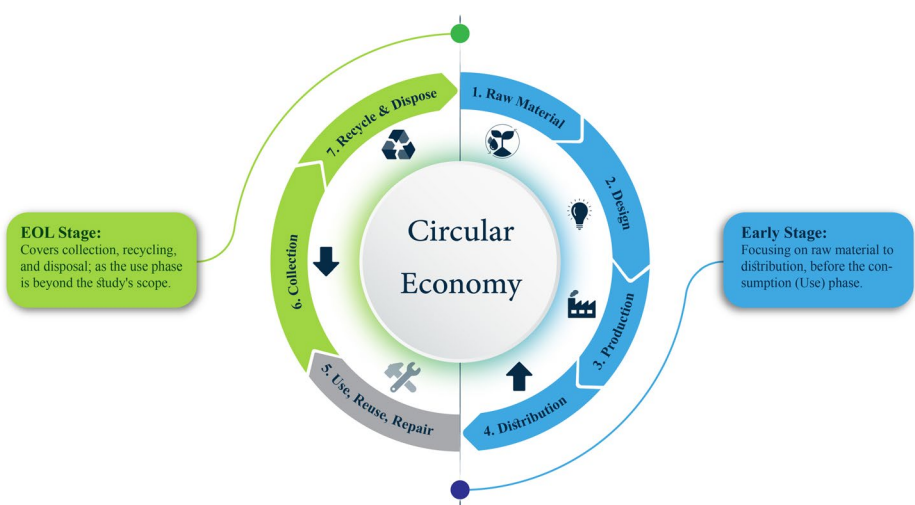


Fig. 2 The CE framework developed for this study is divided into two sections for analysis [30, 55]

the study identifies challenges and opportunities in advancing circular construction systems and contributes to ongoing innovation in sustainable materials.

Material Lifecycle: Early Stages

Raw Material

In the construction sector, non-wood BBMs are utilized for various applications, including interior and exterior finishes, thermal and acoustic insulation, and furniture production [56]. In contrast, wood-based BBMs are predominantly used in structural components while also fulfilling many of the functional roles of non-wood BBMs [57]. Both categories are primarily derived from agricultural by-products and wild plant sources. The dependence of non-wood BBMs on residual biomass streams underscores their compatibility with CE principles, as they enhance resource efficiency by valorizing materials that would otherwise be discarded as waste. Therefore, even at the raw material stage, the process can be considered upcycling, since these inputs are sourced from waste streams and transformed into higher-value construction materials.

Through data extraction and literature review, this study identified a diverse range of wood and non-wood BBMs, categorized in Table 1 by plant type and binder. Binders include mineral, chemical, and natural adhesives, as well as instances where raw plant materials are built without additives, through fasteners or compression.

Table 1 Categorized products of BBMs in construction, classified based on the FAO commodities

Category		Products	Binder	References
Wood	Trees	Lumber (softwood & hard-wood), cork, wastes (e.g., coconut husk, sawdust, wood ashes), etc.	Adhesives, Screws, nuts & bolts, nails, and cement	[57–59]
Non-Wood	Cereals	Maize (corn stalk, corn cob), wheat, rice, hay, rice husk, straw bale	Cement, Concrete, adhesives, clay, no binder (raw)	[60–64]
	Fiber Crops	Hemp, cotton (cotton stalk), flax, kenaf, Jute	Cement, Concrete, adhesives, epoxy, limestone powder, and clay	[23, 61, 65–67]
	Sugar Corps	Bagasse, sugarcane ash	Cement, Concrete, adhesives, epoxy	[61]
	Oilseed Crops	Rapeseed, sunflower (hulls and stalks), lavender, coconut (coir), oil palm leaves,	Cement, Concrete, epoxy	[61, 68–71]
	Root Crops	Potato	Potato starch	[72]
	Fruits and Nuts	Banana, date, hazelnut	Resin and natural binders	[73, 74]
	Fungi	Mycelium	It is a binder itself, which is mixed with other biowaste	[75]
	Animals	Wool, hair, oyster shell	Mortar, clay, no binder (raw), used as an insulator	[76–78]
	Other	Bamboo, bamboo charcoal, reed, cat tail,	Fastener, no binder (raw), Cement, Concrete, adhesives, epoxy	[62, 79]

The diversity presented in Table 1 is mainly attributed to the global availability and widespread cultivation of these plant sources. While some studies, albeit limited in analytical depth, have expressed concerns about the negative impacts of increased BBMs use on land use coverage [80, 81], the availability of bio-sourced material is often underestimated. Approximately half of the world's habitable land is used for agriculture [82], and forests cover about 31% of the total land area, or 38% of habitable land. Additionally, among case studies focused on specific countries, such as France, it has been noted that such sources are abundantly available for construction [21]. Notably, most non-wood BBMs, especially those derived from grasses, originate from agricultural residues. This is because many grass species serve as staple crops essential to human nutrition [83], and consequently dominate global agrarian production [84, 85]. These figures underscore the abundance and geographic dispersion of waste streams, highlighting their potential as a substantial resource base for sustainable BBMs development within the CE framework.

Moreover, although non-wood agricultural plants, particularly staple crops, are widely available globally, a significant portion of their residues and by-products is allocated to alternative applications. For example, a considerable share is used as animal feed [86], with other portions repurposed as fertilizer and animal bedding [87], and some residues converted into biofuel [88], indicating their role across multiple industries beyond construction. In addition, crop residues often serve important agroecological functions when left on the field, such as reducing soil erosion, retaining moisture, and supporting biodiversity and soil health [89, 90]. These ecological roles highlight that diverting residues for BBM production can involve trade-offs that must be weighed against local agricultural and environmental priorities. In contrast, timber is predominantly utilized within construction, accounting for approximately 38% of its global end-use consumption [57]. Notably, in underdeveloped regions, substantial quantities of agricultural waste are frequently burned for land clearing due to limited infrastructure and resource constraints [91, 92], leading to environmental and public health risks while undermining opportunities for upcycling and reuse.

Upon analyzing global agricultural production data by land area and volume from *Our World in Data* [93], shows that many crop species are not regionally abundant enough to serve as practical candidates for bio-based building materials. For instance, wheat is extensively cultivated across many countries, whereas jute is predominantly grown in Bangladesh and India [94]. Similarly, hemp, a promising BM source, faces legal restrictions in some countries because of its classification as a narcotic raw material [95]. In the UK, sunflower farming is minimal [96, 97], with nearly all sunflower products being imported, a supply chain that recent geopolitical conflicts have significantly disrupted [98]. These examples illustrate that regional production patterns, legal constraints, and supply-chain stability all influence the practical availability of bio-based materials and should therefore be considered during material selection when developing CE strategies.

Additionally, unforeseen factors such as pandemics, including COVID-19, have imposed restrictions on the export of global staple crops [99], while geopolitical conflicts further disrupt supply chains that rely heavily on regional stability [100]. Another critical consideration is the variability in plant material costs, which directly impacts the landed cost of commodities. These costs can vary significantly between countries and must be carefully accounted for in any detailed design-phase analysis [101]. Consequently, cost variability was incorporated as a complementary criterion in material selection within the circular economy framework.

Material Design and Selection

To identify gaps in this phase, this study examines whether and how researchers justify material selection with respect to circularity-oriented design considerations. The focus is not on evaluating study outcomes (e.g., mechanical performance, thermal behavior, or GHG emissions), but on analyzing the underlying intentions and rationale guiding material selection. In short, this analysis asks: *Did the researcher explicitly justify why this particular material was chosen, and does that reasoning reflect circular design thinking?*

Although no prior studies in the built environment and construction sectors have provided a comprehensive framework, only two cases were identified that address selected CE criteria. An opinion article providing conceptual guidance on CE [102] establishes standard selection criteria for circular material design, whereas another study [103] proposes value chain selection criteria to identify the most promising bio-based value chains. This research proposes a heuristic framework for BBMs' design and selection, informed by the two aforementioned studies and further substantiated by recent advancements in the literature [31, 38, 81]. Accordingly, six material design and selection criteria (A–F) are introduced to evaluate the extent to which circularity principles have been deliberately integrated into the material selection process.

- A. *General Sustainability Aspects*: Evaluates the material's environmental impact, including resource efficiency, energy consumption, and pollution potential, to gauge the depth of sustainability integration for material design and selection against other criteria.
- B. *Waste-Derived or Renewable Sourcing*: Focuses on materials obtained from waste streams (e.g., agricultural residues, industrial by-products) or from naturally replenishable secondary biomass sources that do not require primary cultivation or compete with food, fibre, or timber production.
- C. *GHG Mitigation and Low-CO₂ Characteristics*: This criterion was used due to BBMs' low-carbon and environmentally friendly characteristics. Evaluates the design or selected material's role in carbon mitigation.
- D. *Recyclability, Biodegradability, and EOL Considerations*: Targeting the renewability and closed-loop of the materials in the EOL scenarios as a design/selection criterion. This element examines the material's potential for reuse, recycling, or biodegradation at the end of its lifecycle.
- E. *Geographical Availability and Supply Chain Security*: Considers the feasibility and sustainability of material sourcing, with emphasis on local availability and continuous supply.
- F. *Special or Context-Specific Characteristics*: Captures properties not covered by other criteria, such as rapid renewability (fast-growing species), low production costs, high durability, or lightweight performance.

Each criterion was converted into a semi-quantitative value using the following scale: 0 indicates that the criterion was not mentioned or considered; 1 denotes partial or implicit consideration; and 2 signifies full and explicit integration of the criterion during the selection or design of BBMs. The "Selection Rationale" section provides an in-depth analysis of the strengths and limitations of each study, particularly regarding the considered criteria. Table 2 evaluates twenty circular BM cases, assessing how material design or selection

Table 2 Presents the material design or selection criteria, specifically for non-wood BBMs. Criteria are as follows: A.General sustainability Aspects, B.Waste-Derived or renewable Sourcing, C.GHG mitigation and Low-CO₂ Characteristics, D.Recyclability, Biodegradability, and EOL Considerations, E.Geographical availability and supply chain Security, F.Special or Context-Specific Characteristics. This table number indicates 0=not considered, 1=partially considered, 2=fully considered

Study	Material		Criteria						Selection Rationale
	Plant	Binder	A	B	C	D	E	F	
Ibrahim et al. [104]	Cotton	Soybean and Resin	2	2	2	0	1	0	Agricultural waste (cotton stalk) was selected for two applications, wood composite building material production and bioethanol generation, supported by LCA. The study highlights cotton's high global availability and yield (1–2 tons/hectare). No end-of-life aspects were discussed.
Keena et al. [105]	Oriented strand board, Cross-laminated timber (CLT), Bamboo, Coconut	Mechanical fasteners	2	1	2	0	2	1	Four wall system scenarios made of CLT, Oriented board, Bamboo, and Coconut were assessed using LCA, with attention to the fast-growing nature and waste potential of selected BBMs. The study briefly addressed supply chain considerations.
Binici et al. [70]	Sunflower & Its Stubble	Epoxy	2	2	0	0	2	0	Sunflower residue and textile waste were mixed with epoxy resin to create 30 × 40 × 2.5 cm rectangular blocks, emphasizing waste valorization. The study identified sunflower as a nationally abundant agricultural by-product in Turkey.
Carcassi et al. [106]	Bamboo particles	Mycelium	2	0	2	1	1	2	Using bamboo and mycelium to create a novel bio-insulation composite for buildings. Bamboo was chosen for its rapid renewability and integration into mycelium-based composites. Although not native to Europe, its potential future availability was acknowledged. EOL was addressed through biogas recovery strategies to avoid landfill or incineration.
Speciale et al. [67]	CLT, grass, jute, and sheep wool	Wood screws and fasteners	2	2	2	0	2	2	A very structured material selection led to three wall options for LCA, consisting of natural stone, cement, CLT as structural, and grass, jute, sheepwool, and rockwool as insulators of walls. Material selection prioritized insulation performance, with specific mention of fast-growing species and regional availability. The study considered multiple CE-related aspects, including CO ₂ footprint, cost, and waste reduction in early design.
Zatta et al. [107]	Timber (Poplar) and Hemp	Synthetic adhesives and Lime	2	0	1	0	1	2	Evaluated the potential of poplar and hemp lime to create poplar plywood elements and hemp blocks. The use of hemp was justified by its fast growth in northeastern Italy. While upscaling and circularity were briefly acknowledged, the study lacked explicit detail regarding CE principles during the selection phase.

Table 2 (continued)

Study	Material		Criteria						Selection Rationale
	Plant	Binder	A	B	C	D	E	F	
Wong et al. [72]	Bamboo	Starch (corn, potato, tapioca), Gelatin, Xanthan gum, Resin	1	1	0	0	0	0	A Netherlands 3D-printed bamboo-based composite case study that could potentially be used for building components and furniture. The study lacked an explanation of circularity criteria or rationale beyond technical feasibility.
Weinland et al. [108]	Rapeseed Straw	Mycelium	2	1	2	0	1	0	Mycelium was combined with shredded rapeseed straw to create insulation panels. While the material's biodegradability was acknowledged, the EOL approach focused on incineration or disposal, with no consideration of reuse or recycling.
Ghisellini et al. [109]	Hemp, straw, bamboo, cork, wool, paper waste	Lime, glue, and compression techniques	2	1	2	0	1	0	This study mapped Italian circular start-ups using locally sourced bio-based materials, such as rice straw, wheat straw, hemp, cork, and wool, for thermal/acoustic insulation, finishes, and non-load-bearing walls, compiling existing LCA data for future assessment but offering no material selection rationale beyond regional availability and no discussion of end-of-life, recyclability, or biodegradability.
Garcia et al. [110]	Agri-waste (not specified), paper, leaves	Tapioca starch	1	1	0	0	0	0	Three bio-based alternatives (paper, leaves, and agri-waste) were assessed for sound and thermal insulation panels. Material choices were presented descriptively without justification linked to circular principles.
Fernandes et al. [111]	Straw, hemp, wool, cork	Not specified	2	1	2	0	0	2	This study compares the life cycle GHG and circular economy performance of low-impact geo- (e.g., rammed earth) and bio-based (e.g., straw, hemp) materials against conventional alternatives as walls and insulation, though it offers limited justification for material selection.
Choi et al. [112]	Coffee Waste and phase change material	Urea-formaldehyde resin	1	2	0	0	1	0	Coffee waste and urea-formaldehyde resin to fabricate interior wall panels, hot-pressed and tested for thermal and acoustic performance. Coffee waste was chosen as a globally available byproduct of coffee consumption, a waste-driven choice aligned with circular economy goals. However, the paper provided no quantitative data on its supply for construction-scale use and did not evaluate or compare alternatives, making the selection appear opportunistic rather than systematic.
Carcassi et al. [73]	Fibers of straw, hemp, kenaf, jute, flax, banana, date palm, wood shaving	Clay, Guar gum	1	2	1	0	0	2	Selected wheat straw over other agri-fibers based on superior printability, ductility, and thermal behavior to create 3D printable earth-fiber composites. Its status as agricultural waste was noted, but no discussion was provided on recyclability, supply chain, or GHG mitigation during the material selection phase.

Table 2 (continued)

Study	Material		Criteria						Selection Rationale
	Plant	Binder	A	B	C	D	E	F	
Can- dido et al. [113]	Straw and Hemp	Mycelium	2	2	2	1	0	1	Mycelium-wood insulation composites were applied in building retrofits to replace traditional insulation. Fast-growing fibers were mentioned, but the study did not elaborate on the rationale for selecting specific plant types or circularity considerations.
Ten- pierik et al. [114]	Sugar beet pulp	Potato starch	1	1	0	0	0	1	Sugar beet pulp bound by starch was selected for its hygrothermal, acoustic, and mechanical properties to create a lightweight building bio-block. The study focused on performance-driven attributes and did not reference CE criteria during material selection.
Go- mez- Cam- pos et al. [115]	Sunflower stalk	Starch	2	2	2	0	0	1	Sunflower stalks were chosen for their waste origin and acoustic performance in insulation panels. The study included LCA to assess GHG impact, but did not elaborate on recyclability or circular design strategy.
Cola- jannni et al. [74]	Dried-fruit shells	Natural binders	2	2	0	2	1	0	The study focused on creating a bio-composite building material by replacing synthetic resins with natural binders to enhance circularity in Sicilian agri-waste composites. Material selection emphasized closed-loop use and extended reuse potential.
Balti et al. [116]	Wood shavings and cotton waste	Gypsum	0	2	1	0	0	0	Wood shavings and cotton waste are bound by gypsum to create building blocks. The study missed key points on circularity and material choice, instead only emphasizing mechanical performance.
Bellei et al. [78]	Oyster shell and cork	Mortar	1	0	1	0	0	0	They developed thermal renders, a type of mortar coating applied to the surface of walls. No explanation was provided regarding material selection, circularity, or design intentions.
Capo- netto et al. [117]	Giant reed	Lime and cement	0	0	0	0	1	1	The final product was a cement-reed block. This study chose giant reed due to the abundance of available studies on straw and hemp. It also stated that giant reed is agro-waste, even though it is a fast-growing wild invasive plant. Nothing was mentioned or considered.

decisions aligned with circularity principles. Each case was examined for the presence or absence of clear justifications tied to the six proposed criteria (A–F). Notably, 18 out of the 20 cases retrieved through the Scopus query explicitly incorporated circularity in their studies. The remaining 12 studies from 32 cases were not suitable for this table.

The evaluation examined 20 recent cases with CE objectives and sustainability outcomes, categorized according to the six material design/selection criteria outlined in Table 2, indicating that nearly all cases acknowledged the sustainability purposes of BBMs and their applications. Furthermore, 12 cases demonstrated strong alignment with sustainability purposes, 6 cases mentioned these purposes, and only 2 instances lacked any explicit reference to sustainability objectives. Of the criteria considered, sourcing materials from waste was the most commonly referenced, mentioned in 7 papers and deliberately implemented

in 9 studies, while the remaining studies did not address it at all. This was followed by the assessment of GHG emissions, which was predominantly addressed through methodologies such as LCA. GHG criteria were purposefully applied in 8 cases and mentioned in 4 others. Unfortunately, criteria D, E, and F within the CE aspect of material design and selection received a significantly higher proportion of superficial consideration rather than in-depth analysis, particularly when compared to the depth of analysis applied to criteria A, B, and C. The least addressed criterion was D, which was mentioned only twice and fully implemented in a single study. This was followed by criterion E, where speculative discussions outweighed purposeful consideration by a ratio of 8 to 3. Lastly, criterion F was addressed on a case-by-case basis across various studies, with 5 mentions and 5 considerations. Notably, none of the D, E, or F criteria surpassed the level of attention given to A, B, and C in terms of their relevance to the CE aspect of material selection and design.

Figure 3 presents the semi-quantitative assessment of the six criteria across twenty studies. Each criterion was scored using a 0–2 scale (0 = not considered, 1 = partially considered, 2 = fully considered). The results reveal a striking neglect of Criteria D and E. Notably, Criterion D appears largely overlooked: it received the lowest possible scores, just 2 for full consideration and 2 for partial consideration, falling below even the lowest scores observed across all other criteria combined. Criterion E reveals a lack of serious engagement among researchers regarding the importance of geographical availability and secure supply chains. Furthermore, partial mentions of this criterion outnumber instances of full consideration or in-depth analysis, making it the only criterion for which this pattern holds. Lastly, criteria A and B appear to be well addressed across the majority of the studies.

This analysis revealed a lack of consensus regarding material design and selection methodologies within the CE context. Additionally, the global dispersion of material availability introduces further complexity, contributing to uncertainties in material design strategies. This challenge is particularly pronounced when attempting to align such strategies with the core principles of CE.

Clustered Bar Chart of Weighted Criteria Scores Across Analyzed Papers

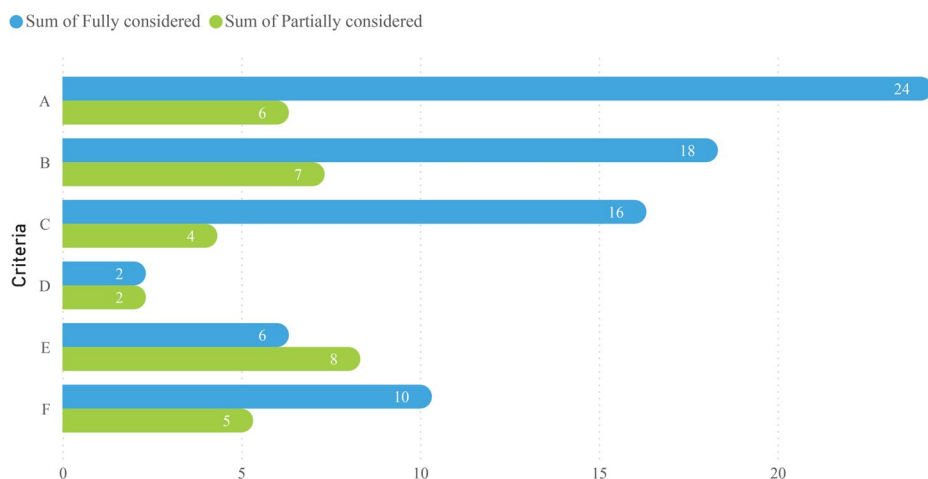


Fig. 3 Weighted criteria scores for material design/selection across twenty recent studies on BBMs and the CE. Scores are based on six CE-related criteria. Blue bars indicate full consideration of a criterion; green bars indicate partial consideration or mere mention

Production and Distribution

Supply chain disruptions can compromise a product's viability by increasing costs, delaying delivery, and reducing customer satisfaction [118], making their resolution vital within the CE. Several studies have recently examined barriers in the production and distribution of BBMs. One study [119] found that no prior research had comprehensively addressed the challenges facing emerging grass-based value chains. Key obstacles included cost constraints, supply fluctuations, uncertainty in agricultural outputs, and a lack of data transparency. Recommendations included fostering collaboration among landowners, enhancing consumer awareness, and setting realistic expectations.

Another study [120] examining bio-based and earthen materials in the construction sectors of Europe and the USA, identified that the limited availability of specialized contractors is the primary barrier to broader adoption. Secondary challenges included a lack of professional expertise, financial constraints, and low awareness of BBMs among construction practitioners. Notably, material availability was regarded as only a slight to moderate barrier; however, the study did not specify whether this referred to limited supply and application or to broader constraints affecting scalability and market potential.

A synthesis of comprehensive review studies that examined the gaps and limitations in the BBMs [21, 36, 121, 122], and previously mentioned studies [119, 120], reveals persistent challenges in BM production and distribution. These include high costs, limited market share, decentralized and fragmented supply chains, restricted raw material sources, insufficient knowledge, low market transparency, inadequate regulatory frameworks, and a shortage of specialized labor. Furthermore, the application of BBMs in the construction sector remains scattered and inconsistent. Furthermore, the application of BBMs in the construction sector remains scattered and inconsistent [123], highlighting the market's heterogeneous and underdeveloped nature. Despite longstanding recognition of the challenges in BM production and distribution, and earlier recommendations, effective interventions remain notably lacking.

Many of these challenges originate in the design phase, where key CE considerations are frequently overlooked. As demonstrated in Sect. 2.2 and Fig. 3, similar oversights in material selection and design recur across the production and distribution phases, particularly in parameters D, E, and F. These omissions have led to inefficiencies in supply chain performance, recyclability, EOL management, and cost optimization. A primary cause is the disproportionate focus on LCA and GHG mitigation by material designers. While carbon reduction is critical, this narrow emphasis often neglects equally vital aspects such as supply chain resilience and material circularity. This imbalance reveals a significant gap in CE discourse and underscores the need for a more integrated, framework-based approach to material design that supports truly sustainable circular practices.

Material Lifecycle: End-of-Life Stages

Collection

Collection refers to the retrieval of materials at the end of a building's service life, involving demolition and the transportation of retrieved BBMs for recycling, reuse, or disposal.

The ease of retrieval is influenced by both the material type and construction method; For instance, hempcrete, typically used in block-type wall systems, is considerably more challenging to reclaim than light steel frame systems [34]. A widely recognized approach to enhance retrieval is design for disassembly (DfD), which emerged in modern construction during the early 1990s [124] and supports waste minimization and circularity [34]. Since 2015, DfD has attracted increasing attention in steel and concrete structures, with a notable rise in timber applications observed in 2021 [124]. In contrast to timber and conventional practices, the initial screening of BBMs highlights a lack of robust data on their recoverability, raising critical concerns regarding their collection and integration into circular practices.

The most recent example of DfD in BBMs is found in prefabricated straw bale systems, which contribute to reduced CDW and accelerate construction timelines [125, 126]. Similarly, load-bearing prefabricated hempcrete exterior walls have been observed in commercially available products. Both systems can be easily disassembled at end-of-life for reuse or recycling.

Unfortunately, no comparable data were found for other BBMs, revealing a critical gap in the literature regarding their integration into CE practices. Despite limited data, prior practices with block-shaped BBMs, such as clay fiber mix, bagasse, and hemp blocks, can be analogized to conventional wall blocks like autoclaved aerated concrete and clay bricks, suggesting that recovering these materials poses considerable, if not greater, challenges. Their inherent material composition and the bonding binders used in laying make them significantly more challenging to collect and process at EOL than modular systems, specifically DfD.

It is essential to recognize that BBMs are inherently susceptible to degradation, a process that, while unavoidable, can be substantially delayed [127]. This characteristic may affect the recoverability of unprocessed materials such as straw bales and wool; however, evidence on this issue remains limited and warrants further investigation. Moreover, a recent study found no documentation of dedicated facilities, collection points, or centralized depots for the construction sector [128], particularly concerning the EOL management of BBMs, where the market infrastructure remains underdeveloped [129]. This underscores the urgent need for improved logistics and procurement in the BM collection stage.

Recycling and Disposal

BBMs have been reported to undergo post-consumer processing pathways [27]; however, these pathways are often insufficiently specified. As discussed in Sect. 4.2.1 and 4.2.2, and supported by the author's observations, binders play a significant role in determining the recyclability and circularity potential of each material. BBMs can be broadly categorized into two groups: unprocessed materials and processed composites that incorporate binders. To support this classification, Table 1 outlines three main types of binders commonly used in BBMs: (1) synthetic and mineral binders, such as cement, lime-based compounds, and resins; (2) natural binders, including bio-based options like gum arabic and mineral binders such as clay; and (3) mechanical fasteners, where materials are used in their raw form without added binders. This framework enhances the analysis by clarifying distinctions between material types. The circularity pathways of both processed and unprocessed materials are illustrated in Fig. 4, providing a visual representation of their roles within CE strategies.

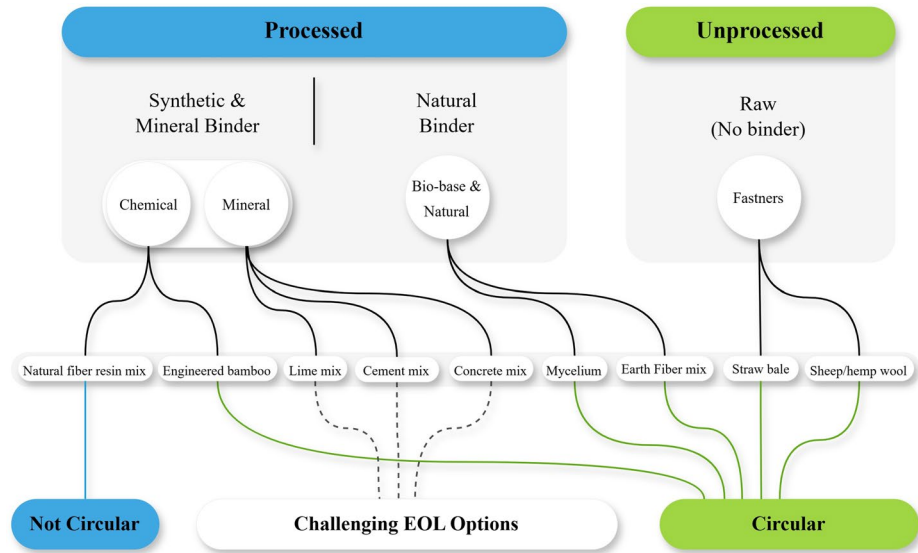


Fig. 4 Circularity of the assessed materials in the literature is illustrated across defined categories based on processing level and binder usage

Processed Materials

Processed materials undergo extensive preparation and are often combined with mineral or chemical adhesives such as gypsum, lime, and cement, resulting in composites like hempcrete, fiber-reinforced concrete, and wood wool cement boards. They not only exhibit a low carbon footprint due to the sustainable cultivation of hemp, but also sequester additional carbon throughout their lifecycle [130]. Unfortunately, despite extensive research, the EOL stage, degradability of cement/lime materials such as hempcrete in landfill conditions, and potential applications of recycled hempcrete remain insufficiently explored [131]. The primary method for recycling mineral-bonded materials like hemp lime involves crushing them into aggregates for reuse in new hempcrete production [132]. Another study on vegetal concrete confirms the viability of recycling BBMs for new construction applications, though residual lime in recycled straw aggregates significantly influences biocomposite performance [133]. Nonetheless, existing literature recommends the challenging approach of aggregate recycling for lime-based processed materials, yet offers no evidence supporting upcycling or complete biodegradation pathways. Furthermore, the logistics and practical feasibility of the recycling process, including collection, storage, and supply chain management, remain unclear and require further investigation.

In the context of resin-bonded materials, they are inherently challenging to recycle, reprocess, and solubilize due to their permanently crosslinked structures formed after curing, thereby reducing their environmental sustainability [134, 135]. Although resin-bonded non-wood BBMs have been examined in previous studies [70, 104, 107], neither these nor any other existing study has addressed their recyclability, upcycling potential, or biodegradability.

One material that has partially incorporated recycling and renovation considerations during the design phase analysis is mycelium, which was utilized as an insulating material. Initially introduced in 2007 for sustainable packaging applications [136], then gained traction in the construction industry around 2012 due to its promising thermal performance and insulating properties [108]. It functions as a bio-based binder, demonstrating considerable success in construction applications and showing strong potential for future use due to its biodegradability [137], which makes it compatible with CE goals. Overall, Mycelium is considered a key BBMs, encompassing nearly all aspects of the circular economy, as highlighted in studies such as [113]. This natural binder can be endlessly recycled, is biodegradable, and can recycle agri-waste as well [108, 138]. It is primarily used as an insulator and has a lower environmental impact compared to mineral and synthetic alternatives, although its production depends on the availability of agricultural waste and resources [108]. Recent studies highlight that while mycelium insulators have approximately half the overall environmental impact of conventional insulators, they also exhibit higher land and water use, increased electricity consumption, and greater territorial toxicity, limitations that require further investigation [39, 139]. Interestingly, the previously mentioned study [39] ultimately recommended fast-growing fibers over hemp. However, this recommendation was made after the study concluded, displaying the lack of a well-defined material selection or design process. Such a critical gap and oversight that, based on Fig. 3, is common and can negatively affect the circularity of materials from the design stage by scholars, designers, etc. In contrast, our study addresses this gap by emphasizing the need for clear material selection criteria within the CE framework, thereby helping to overcome such limitations.

Among promising emerging circular BBMs, fiber-reinforced (3D-printed) earth materials demonstrate strong potential due to their complete recyclability and reusability [140, 141]. Conversely, the addition of cement to earth-based materials compromises their recyclability, preventing direct reuse in new construction applications [140]. Comparable earth-based binders in BBMs include light straw clay blocks and cast-in-place (poured) applications, both recognized for their local availability as well as their complete recyclability and biodegradability, leaving no harmful or persistent environmental residues [142]. In contrast, conventional clay bricks are only partially recyclable; reuse is limited by contamination and damage, recycling is energy-intensive, and landfilling is common due to sorting difficulties and economic constraints, issues similar to those faced by lime-based materials [143].

Bamboo, a fast-growing grass that regenerates without replanting and matures within 3–5 years, offers timber-like properties and holds strong potential for CE applications, particularly in recycling and sustainable disposal [144]. In the construction industry, bamboo is primarily used in the form of sheets and boards for finishing, cladding, and flooring. However, its use as a structural load-bearing component in large-scale buildings has only recently gained attention [145]. Structural bamboo is produced through lamination, resulting in mechanical properties and shapes comparable to glued laminated timber (GLT) and laminated veneer lumber (LVL) [146]. While the production of durable bamboo materials typically involves resins and adhesives, factors that significantly impact their circularity, they are still considered a more sustainable and circular alternative to abiotic construction materials [144]. This is due to its similarity to timber, bamboo can be easily disassembled, sliced, and reused at its EOL, further reinforcing its potential within CE frameworks. A relatable example includes the recycling of scaffolding into furniture-grade materials [147].

Notably, no additional studies were identified, underscoring the need for future research on the recycling and upcycling potential of engineered bamboo structures and components.

Unprocessed Materials

While the use of raw BBMs in construction remains limited and relatively obscure [142], recent studies indicate a growing public interest in their potential as sustainable alternatives [148, 149]. Among these materials, straw bale is widely recognized for its thermal performance and is employed in both load-bearing and non-load-bearing applications [150]. However, its broader adoption is constrained by factors such as moisture sensitivity, vulnerability to pests, lack of regulatory support, and insufficient skilled labor [151]. Its limitations extend across research, design, manufacturing, policy, and public perception [150], echoing challenges faced by other BBMs discussed in Sect. 3.3. Moreover, although they are fully biodegradable [152], straw bales are susceptible to gradual degradation over time [153], raising concerns as reclaiming these raw materials, making the reclamation of non-prefabricated forms particularly difficult [125]. Lastly, potential recycling options include conversion into wood composite boards, biogas, or biomass energy [125].

Raw bio-insulators such as sheep and hemp wool have garnered attention for their recyclability, environmental benefits, and thermal performance comparable to conventional mineral-based insulators [154]. Similarly, they demonstrate equivalent properties to conventional insulators [155]. However, both face persistent challenges, including poor fire resistance and significantly higher costs [156, 157]. Such factors have limited the market penetration of bio-based insulators in the EU, where they represent only 1.6% of the market, compared to 22% for mineral wools [129]. Additionally, unlike the sheep or hemp wools, mineral wools are fully and infinitely recyclable into new products [158], thereby aligning more effectively with closed-loop CE practices.

Overall, raw BBMs generally exhibit limited practical applications, with scarce supporting data. Even when identified, such as shredded paper, sugarcane waste, and wood pulp or particles, their utility is predominantly restricted to processed forms. These materials consistently lack both robust application evidence and sufficient data to support their integration into the CE framework.

Discussion

Limitations of the Study

The use, reuse, repair, and maintenance phases were excluded due to the saturation of performance testing studies and the wide variety of BBMs and configurations, which limited generalizability.

Research Gaps

Previous studies employed systematic reviews and structured questionnaires, which were unable to capture the depth and diverse scope of BBMs within the CE. In contrast, this research uses a semi-systematic review approach, providing greater flexibility to identify

evident gaps and limitations. This may be attributed to previous studies having underestimated the broadness of BBMs' scope and the key factors influencing their circularity.

The analysis in Sect. 3 highlights the gaps and limitations of BBMs within the CE framework. The raw material stage exhibited the least significant gaps and limitations, whereas the subsequent stages revealed numerous critical issues. It is likely that the absence of intentional material design and selection strategies has significantly impacted the production and distribution phases. Overall, the EOL phase is often neglected in the initial design and production of BBMs (e.g., Fig. 3), which hinders the development of truly circular, recyclable materials. Although the production and distribution phases are extensively explored, a pressing question remains: *Why have these well-documented issues not been addressed by researchers or industry entities?*

Moreover, Sect. 4 revealed that the EOL phase has not been adequately considered or analyzed in relation to circularity. No strategies, such as DfD, were integrated during the design phase to facilitate material recovery, nor were any deposit schemes or mechanisms proposed to support the reuse, recycling, or disposal of BM components. Based on our observations and the findings of [144], the commonly used binders significantly influence the recyclability and biodegradability of BBMs, which directly affects the circularity. Furthermore, the literature on EOL scenarios for various BBMs was notably scarce, making it difficult to provide substantial analysis and resulting in limited coverage of the topic. Interestingly, while the raw material stage indicates sufficient availability, the production and distribution stage reflects a contrasting scenario. This disparity may be attributed to inadequate sourcing and supply chain management, as the origin of BBMs is often region-specific, and there is a lack of intermediaries to connect suppliers with consumers in neighboring areas effectively. Nonetheless, most BBMs demonstrated promising performance during the reuse, recycling, and disposal phases. Among them, two non-wood BBMs, mycelium and bamboo, stood out for their exceptional circular potential. Mycelium can bind various biowastes to create a fully recyclable and biodegradable material; however, it is typically produced in block form, and the interlayer binding materials used can significantly influence its overall circularity. Bamboo, a fast-growing grass with timber-like properties, presents multiple EOL scenarios, including upcycling, recycling, and biodegradation, depending on the treatment applied. Another notable material is 3D-printed fiber-reinforced earth composites, which exhibit complete recyclability and reusability, making them a strong candidate for future circular applications.

Fundamentally, the concept of CE encompasses a broad and context-dependent meaning that cannot be captured by a single definition [38]. However, drawing from prior research and prevailing interpretations, CE is often characterized at the macro level by the 4R framework, reduce, reuse, recycle, and recover, emphasizing sustainability, waste minimization, and the establishment of closed-loop systems [29, 37, 38, 54, 159]. Unfortunately, while studies highlighted in Table 2 and many more [21, 36, 119–121] have evaluated the CE's early stages, they have neglected the fundamental aspect of CE, and the current literature on definition appears to align with the CBE [160] more than CE. If the focus is solely on the CE and disregards sustainable impacts, glass and ferrous & non-ferrous metals waste emerge as the most circular materials, as they can be infinitely recycled without losing their properties and exhibit the highest recovery rates, making them ideal for achieving CE principles [161, 162]. Therefore, upcycling the biowaste to waste-derived BBMs alone does not inherently qualify them as circular materials, as the EOL scenarios of BBMs are still uncertain. This

gap underscores the need for a systemic shift toward CE principles in BM design. Table 3 depicts the summary of the BBMs' gaps within all CE stages. The barriers summarised in this table were identified across the full semi-systematic evidence base, including the 32 studies from the structured review and additional sources from narrative synthesis and snowballing.

Table 3 Summary of gaps and limitations of BBMs through the CE stages

	Phases	Gaps	References
Early Stages	 Raw Material	• Diversity and decentralized sources of raw materials present both opportunities and challenges. • Restrictions of some sources within certain regions due to climatic conditions and regulatory barriers. • Probability of temporary raw source disruptions during global conflicts and pandemics that can affect circularity which can affect circularity.	[94–99]
	 Design & Selection	• Absence of any design criteria or framework for materials design for CE. • Lack of attention during the design phase to the former phase and the EOL phase.	[39], Table 2, Fig. 3
	 Production & Distribution	• High production costs • Limited market share • Decentralized market • Limited raw material sources • Complex supply chain • Lack of public and expert knowledge • Poor market transparency issues • Inadequate regulations • Specialized labor shortage • Inconsistent application across markets	[21, 33, 35, 36, 118–123]
EOL Stages	 Collection	• Lack of DfD considerations, stemming from the design phase. • Inadequate infrastructure for material collection and no available centralized depot for storage, reuse, and resale. • Degradation and disintegration of BBMs make them challenging to retrieve.	[127–129]
	 Recycling & Disposal	• Many recycling challenges stem from early design-phase neglect and the lack of systematic CE-based material selection criteria. • Proposed EOL strategies are often challenging or unfeasible. • Synthetic and mineral binders significantly affect the circularity of materials. • Reuse, recycling, and upcycling pathways of many BBMs are underexplored, especially for lime and resin-bonded BBMs.	[131, 144], Section 4.2

Comparison to Timber

A brief comparison of timber and non-wood BBMs was conducted to highlight the findings, overlapping gaps, and future prospects. Timber was included in this comparison because it aligns well with circular economy principles, given its ability to be disassembled, reused, and adapted across various applications, with components recoverable through deconstruction and repurposable in new construction or as alternative wood-based products.

From the perspective of raw material design and selection, wood and non-wood BBMs exhibit diverse forms and species, yet they serve distinct purposes. Non-wood BBMs, often derived from waste and processed into particulate forms, while wood BBMs are typically logged in a solid form. Although agricultural waste is widely available globally, its utilization in the construction industry remains limited, with the majority directed toward biofuel and feedstock production, while wood and timber, despite their limited accessibility, are predominantly used in construction applications worldwide [57, 86–88, 163, 164]. Regarding wood availability, forests cover nearly a third of our land area, which is the primary source of wood [82, 163]. Almost half of this forest cover is concentrated in Russia, Brazil, Canada, and the United States [164], making wood availability regionally constrained. In contrast to widely available non-wood agricultural plants, wood resources are notably scarce in regions such as the Middle East and Northern Africa, posing significant challenges for their accessibility and utilization [164].

From a design aspect, non-wood BBMs have encountered some significant challenges and revealed no integrated approach, specific design, or selection criteria. In contrast, the challenges associated with the characterization, processing, and design of timber and mass timber have been effectively addressed [165]. Their use and considerations are also different; due to the non-wood BBMs' limited mechanical strength, they are rarely utilized in the construction of load-bearing structural members [166], whereas wood and timber are utilized not only in low-rise residential buildings, advancements in wood technology have now enabled the construction of mid-rise and high-rise structures [57, 167]. Therefore, there are some distinct applications in source, design, and materials selection, which appear to be due to the known wood market and products, unlike the decentralized and fragmented non-wood BBMs.

Most non-wood BBMs lack standardized codes, widespread public awareness, and a robust network of specialized contractors, which limits their scalability and global adoption. Except for bamboo, which shares many similarities with timber and has gained increasing international recognition, including the recent introduction of structural standards such as ISO 22,156 and ISO 19,624 [145, 168]. In contrast, timber and engineered wood products benefit from a mature circular infrastructure. Several studies have analyzed EOL scenarios for wood-based BBMs through case studies, developed frameworks for integrating circularity into life cycle assessments, and identified critical areas for further research [169–171]. Timber's established status, supported by codes, familiarity, and skilled labor, enables greater reuse, recycling, and upcycling than non-wood materials.

Numerous studies have assessed various EOL phase scenarios in the environmental evaluation of wood, encompassing recycling, disposal, and reuse [172–174], alongside advancements in recycling techniques and processes for wood and timber as a sustainable alternative to virgin timber [175]. For instance, a study assessing Quebec's construction and demolition waste, which contributes 13% of Canada's total wood waste from such sources,

with its potential to be upcycled into biochar, aims to transform this waste into a high-value product [170]. Another example of pursuing a circular economy in the timber industry involves redirecting wood construction waste from the furnace and disposal to a 3D-printed Biowall, effectively utilizing an upcycling process [176]. Moreover, timber-based structures and engineered wood products, such as CLT, GLT, and LVL, have demonstrated greater popularity over their concrete counterparts within the DfD scope [124], which is considered a path to achieving circularity [177]. It is worth noting that although engineered timber products and processed bamboo are not entirely circular, they can still be considered as contributing to circularity.

Overall, wood-based BBMs demonstrated a well-established degree of circularity, whereas non-wood BBMs exhibited significant deficiencies across nearly all stages of CE. While many challenges related to material characterization, processing, and design for whole timber construction have been substantially addressed, and mass timber is expected to complement other timber products in global construction over the coming decades, it cannot serve as the sole material source. Exclusive reliance on wood in construction may result in increased net emissions by 2050 [178] and contribute to biodiversity loss [179], primarily due to intensified logging practices and the conversion of natural ecosystems into monoculture timber plantations. Notably, a Danish case study [180] demonstrated that combining fast-growing BBMs with timber structures can effectively reduce land-use impacts, offering a viable solution for circular and self-sufficient construction. Therefore, integrating non-wood BBMs alongside wood-based alternatives is essential to enhancing circularity while avoiding overreliance on a single natural resource.

Future Prospects

A recent study [128] recommends a complementary CDW policy mix, including economic, technical, operational, and recovery regulations, to enhance circularity. Although operational instruments, referred to as practical measures and tools that support the on-site implementation of waste management policies, are the least adopted, they show the strongest association with higher recovery rates. The study also notes that no reverse logistics infrastructure for storing and reselling used CDW materials was identified.

Another study [181] recommends a similar CDW policy mix of local regulatory frameworks, enforcement mechanisms, infrastructure capacity, and construction practices. Building on this, achieving a CE and ensuring policy compliance requires active engagement at the local level, supported by industry-wide efforts to enhance cooperation and standardization across both digital and physical markets for secondary materials. Therefore, a unified, operational platform, guided by an adaptive coordination network grounded in local policies, is essential to enable stakeholder engagement and informed decision-making, both critical for a successful global circular economy transition.

To reach CDW recovery and circularity, prior research highlighted the need for operational and industry collaboration to develop localized solutions, particularly for BBMs, which are associated with heterogeneous and fragmented products and present supply chain issues throughout the CE stages. Applicable solutions addressing construction circularity limitations have been demonstrated across various industries and business models. For instance, intermediaries have been widely employed to mitigate issues related to decentralization, supply chains, costs, and other systemic challenges [182]. Intermediaries are enti-

ties that improve consumer search efficiency by providing a search platform and typically possess greater expertise in local regulations and cultural practices, and frequently bear formal responsibility for the product [183, 184]. Furthermore, they manage their supply chain, indirectly make regulations, conduct market analysis, and strategize future development initiatives and innovations [185, 186].

A notable example of an intermediary practice applicable to non-wood BBMs is the circular timber hub, which collects, stores, and redistributes construction waste for secondary use [187]. Owing to their effectiveness within the CE framework, several timber-based intermediaries are analyzed as successful CE models. One prominent case in the construction sector, particularly for wood-based BBMs, is Timber Development UK [188], which functions as a comprehensive platform offering a searchable database of suppliers, designers, and builders, along with market data, educational tools, a technical library, and industry events. It also offers expertise on local building codes, standards, and sustainability practices, supporting regulatory compliance and the adoption of best practices. As a key authority, it facilitates connections with other hubs and promotes the responsible sourcing and use of timber, with an emphasis on quality, safety, and sustainability. Similar platforms provide additional services, including manufacturer and supplier directories, technical drawings and specifications for fire safety, acoustics, and design [189]; guidance on net-zero standards and biogenic carbon accounting [190]; as well as a range of publications, educational resources, and advisory services [191]. Overall, despite growing attention to timber and engineered wood products, many non-wood BBMs still lack structured support through dedicated intermediaries. This absence highlights a significant gap in both market infrastructure and coordination.

Based on these findings, a viable pathway forward is the establishment of a dedicated, regional centralized hub for BBMs, distinct from the general uncoordinated local circular construction hubs proposed in prior studies [177, 192, 193], which primarily address deconstruction or function as digital platforms. This distinction is necessary due to the logistical, heterogeneous, and underdeveloped nature of the BBMs market, as well as the need to facilitate their mainstream integration into the construction sector.

Such a hub would facilitate the sale and resale of materials, standardize labeling practices, develop verification protocols, manage inventories, and offer collection and demolition services. Additionally, it would disseminate educational resources and provide analytical data to assess the availability of suppliers and contractors. For example, the hub could incorporate material passports to enhance transparency and stakeholder confidence by offering centralized, unbiased verification of key attributes, such as fire resistance, recyclability, assembly and disassembly details, technical specifications, quality, and pricing. These digital documents provide information on recyclability, physical and thermal properties, toxicity, and carbon footprint. Furthermore, these hubs can function as adaptive frameworks that align with various materials, local codes, construction methods, and resource availability. For example, while straw bale or hempcrete may be viable in certain climates, bamboo or mycelium-based composites might be more suitable in others with specific minimum hygro-thermal requirements.

The regional central hub could also serve as an incubator or accelerator for startups developing novel BBMs or construction methods. By offering exposure, market entry support, and access to data, it would stimulate innovation and attract venture capital aligned with circular construction principles. Notably, future success will also depend on cultural

shifts and knowledge dissemination. The proposed hub could provide training modules, vocational education, and certification programs to equip designers, contractors, and regulators with the competencies required to handle BBMs responsibly. Without parallel investments in education and behavioral change, the operationalization of circular models will remain limited.

Considering the uneven distribution of waste generation and the diversity of management infrastructures across regions, a regionally tailored approach may offer greater efficacy. For example, distinct systems could be developed within the European Union and North America to align with their respective regulatory frameworks, resource availability, and market dynamics. Such a regionalized yet centrally coordinated framework could enhance market accessibility by fostering interoperability among local systems. Moreover, it would address existing challenges related to CDW collection, cross-regional redistribution, and cost inefficiencies. By streamlining processes and enabling more consistent regulatory oversight, this model holds significant potential for the effective integration and upscaling of BBMs within the construction industry.

With the BBMs market's gradual expansion, a regional central hub could become instrumental in overcoming long-standing material selection and design difficulties that researchers and industry professionals frequently encounter. Ensuring the hub's effectiveness and financial sustainability may require operating as a for-profit entity. This could involve models such as charging membership fees, forming public–private partnerships, applying transaction-based fees, or offering subscription-based access to data and insights, each with a track record of scalability and long-term viability. Such models offer greater long-term viability compared to time-constrained, policy-driven initiatives like the RECONSTRUCT EU Project and the Circular Cities and Regions Initiative, among many similar ones, which are inherently limited by the temporary nature of funding mechanisms such as Horizon Europe. Additionally, the capacity to establish and sustain similar hubs is not universally available, particularly outside the EU.

Figure 5 outlines the core functions of a proposed regional centralized hub, derived from intermediary-focused studies, to address circularity challenges in BBMs. The functions are categorized into four domains: Engagement & Training, Supply & Logistics, Standards & Codes, and Data & Innovation.

This study offers valuable insights for policymakers, private stakeholders, and both industry and academic professionals by identifying critical gaps, strategies, and material selection criteria that can inform the development of future circular hubs and regulatory frameworks. A relevant example at the policy level is the recently established temporary Reconstruction EU Project, which aims to foster circularity by establishing a European circular construction hub. While commendable in its ambition, the initiative adopts a broad, multi-material approach that spans established markets such as steel and cement. However, these industries often possess entrenched stakeholders with limited incentives to adopt circular business models that may reduce demand for their primary products. As such, the project's success could be constrained by conflicting market interests and stakeholder resistance.

In contrast, this paper argues that the heterogeneous and underdeveloped nature of the BBMs market presents a unique opportunity to formulate more actionable and inclusive strategies for engaging regional stakeholders across one or multiple regions. By targeting an emergent sector with fewer structural barriers and greater flexibility, a BM-specific circular hub, such as the model proposed in this study, can foster early-stage collaboration, shape



Fig. 5 Key responsibilities of a regional centralized circular hub for BBMs, organized into four strategic domains

future standards, and accelerate circular adoption. In doing so, the insights and framework provided herein may serve as a valuable resource for initiatives like Reconstruction EU, Circular Cities and Regions Initiative, and similar industry entities, offering a pathway to more targeted, scalable, and stakeholder-aligned strategies.

Finally, regarding the material selection and design framework for circular BBMs, this study revises the first criterion (general sustainability for benchmarking against other materials), as presented in Sect. 3.2, and replaces it with DfD, given its critical role within the CE model. To assist readers in applying these criteria, the assessment approach, and the referenced resources, Table 4 is provided. These criteria are not weighted or prioritized, as the framework is intended as a heuristic synthesis rather than a decision-ranking tool. The examples provided below are resources commonly used in the literature to support future studies seeking to operationalise the criteria. Figure 6 presents a summary of the revised framework, which is intended to guide future research and applications.

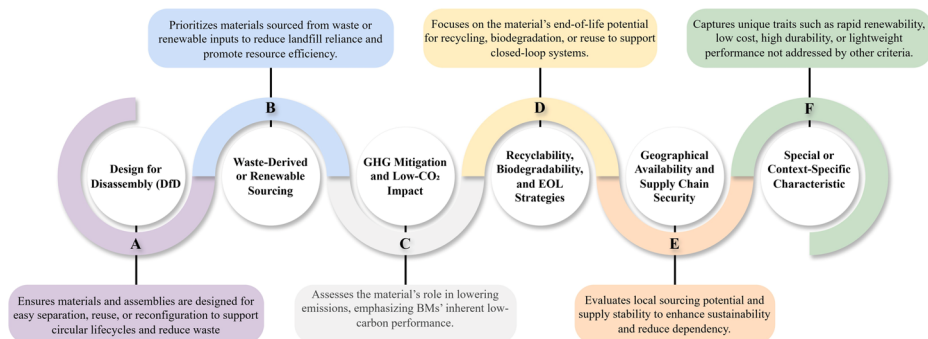
Conclusion

This review identified 22 barriers across all CE phases. While raw material sourcing poses relatively few challenges, the production and distribution stages reveal persistent gaps, and substantial obstacles remain across the later stages of circularity. These limitations are primarily rooted in the absence of circular thinking at the design stage. For instance, materials sourcing and availability, DfD, and binder types, whether natural, synthetic, or omitted, have a decisive impact on recyclability, yet are seldom considered in design and research. Although BBMs align closely with the principles of the CBE due to their renewable, regenerative, and low-impact characteristics, their integration into CE systems remains conceptually fragmented and practically inconsistent. Current circular approaches in the BM domain often lack coherence and deviate from core CE objectives.

To identify design gaps, this study employs a six-criteria material selection framework that can serve as a foundation for future material design in circular construction. The criteria of sustainability, waste origin, carbon mitigation, recyclability, availability, and functional

Table 4 Assessment approach and references to conduct the material selection/design process

Criterion	Example assessment approaches	Reference/resource types
A. DfD	DfD guidelines and designs	Design for disassembly and adaptability (DfD/A) in sustainable building and civil engineering works (ISO 20887), tested design cases,
B. Waste-derived or regeneratively sourced by-products	Waste hierarchy frameworks, material flow analysis, and second stream resources	Material flow analysis, residue availability studies, and by-product assessments.
C. GHG mitigation/low CO ₂	Carbon footprint calculation	Carbon accounting, embodied carbon guidelines (EN 15978), LCA frameworks (ISO 14040/44), and environmental footprint indicators. Ecoinvent, ICE database
D. Recyclability, biodegradability, EOL	Recyclability and reuse tests/Biological degradation test	Circularity indicator frameworks and compostability standards. Considering not using Mineral binders. Reading binders' biodegradability and characteristics.
E. Geographical availability/supply chain	Supply-chain mapping	FAO statistics, national/regional agriculture and forestry database
F. Context-specific characteristics	Material property testing	Material property tests, product-standard documents, and mechanical performance literature

**Fig. 6** Design and selection framework for circular BBMs, highlighting six key criteria to support CE principles

properties offer a replicable and rational basis for guiding circular material decisions across both academic and policy contexts. Following the final analysis, the integration of DfD is strongly recommended as a key component of the proposed framework. The results highlight mycelium and bamboo as notable exceptions that embody CE principles, alongside

fiber-reinforced earth materials. Mycelium, in particular, demonstrates strong closed-loop potential due to its biodegradability, self-binding characteristics, and compatibility with agricultural waste. These underutilized materials warrant greater attention in future design innovation and R&D investment.

For future prospects, this study proposes the establishment of intermediaries operating as multifunctional, regionally centralized hubs for BBMs. Structured as for-profit entities to ensure long-term viability, these hubs would address the fragmentation of the BM sector by centralizing material data, providing regulatory and design support, facilitating contractor collaboration, and enabling reverse logistics. Their role would be to coordinate fragmented industry practices, enhance market transparency, and accelerate the transition from theoretical circularity to actionable implementation.

Additionally, a balanced use of both wood and non-wood BBMs is strongly recommended. Overreliance on timber risks deforestation and biodiversity loss, while exclusive use of non-wood BBMs lacks scalability. While timber offers structural integrity and standardization, non-wood BBMs contribute thermal, acoustic, and ecological benefits. The integration of their complementary strengths provides a more resilient and cohesive pathway toward realizing circularity and achieving a sustainable built environment.

In sum, while bio-based materials hold immense promise for enabling a circular built environment, this potential can only be realized through systemic coordination, intentional design strategies, and regionally tailored infrastructure. Advancing from theoretical sustainability to practical circularity requires not only better materials, but also better systems, frameworks, and a unified commitment across the entire construction value chain.

Future Research Directions

1. *Expanding CE Lifecycle Phases:* This study focused primarily on the production and end-of-life phases of BBMs within the CE. Future research should investigate the underexplored phases of *use*, *repair*, and *reuse*, which are highly context-dependent. Key areas include strategies to extend the in-use lifespan of batteries and enable user-driven maintenance, as well as reparability practices and renovation.
2. *Operationalizing the Intermediary Hub Model:* The proposed centralized intermediary hub offers a conceptual framework for integrating circular practices in the construction sector. Future work should assess its feasibility through empirical studies in different regional and regulatory contexts. Research should address challenges in circular material selection, supply chain logistics, labor coordination, regulatory compliance, and infrastructure for collection and redistribution. Additionally, financial models, stakeholder governance mechanisms, and digital tools, such as material passports and circularity databases, require further development to support practical implementation.

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Declarations

Ethics Approval Not applicable.

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